D06 · Freight Management & Mode Selection

01 · Transportation mode selection: road, rail, air & ocean

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1 WHAT IT IS

Mode selection is the decision of how each lane moves — road (FTL/LTL), rail or intermodal, air, or ocean — based on the product's value density, the service the customer was promised, and the total landed cost, not the freight rate alone.

2 WHY IT MATTERS

Freight rate is the visible cost; inventory in transit, safety stock and lost sales are the invisible ones. **A slower, cheaper mode is only cheaper if the inventory it adds costs less than the freight it saves** — the classic transportation–inventory trade-off **(Chopra and Meindl — Supply Chain Management, textbook)**. Companies that choose mode by rate alone systematically over-use air for emergencies and under-use rail and intermodal for stable, heavy flows.

3 HOW IT IS DONE

1. **Segment lanes by value density and service need.** High-value, urgent flows tolerate premium modes; heavy, predictable flows are candidates for rail or intermodal.
2. **Compute total landed cost per mode.** Freight + in-transit inventory carrying cost + safety stock for transit variability + handling and drayage.
3. **Weigh reliability, not just average transit time.** Transit variability drives safety stock; a slower but predictable mode can beat a faster erratic one.
4. **Review the mode mix at least annually.** Rates, volumes and network change; yesterday's optimal mix drifts. Track premium-freight spend as the early warning.

IN PRACTICE

Always run the full TCO, never freight price alone

The visible freight line is frequently the smaller part of the true cost for high-value products.

Re-evaluate mode strategy after every major nearshoring or product-value shift

Mode decisions made under Asia-centric lead times become wrong the moment production moves to Mexico or product ASP rises significantly.

Use air strategically for NPI and high-obsolescence windows

Air at launch -> ocean as the product matures is a high-ROI dynamic mode strategy.

ILLUSTRATIVE CASE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case — High-value electronics components from Asia — Product value $420/kg, 45-day ocean transit, 10% WACC.

TCO Component	Ocean	Air
Freight	$0.95/kg	$5.13/kg
In-transit inventory cost	$5.18/kg	$0.35/kg
Total TCO	$6.13/kg	$5.48/kg

Result: Air is $0.65/kg cheaper on TCO. At 2.4 M kg/year the annual advantage is $1.56 M. For standard components at $18/kg the ranking reverses completely.

HOW IT IS MEASURED

Transportation Cost per Unit (full TCO)

Freight + in-transit inventory cost + insurance / Units moved.

Benchmark: Track trend by mode and corridor; absolute number is less useful than directional change after mode shifts.

Modal Mix %

% of volume (tons or value) by mode.

Benchmark: Optimal mix is the output of a TCO model, not an industry average.

WHAT YOU WOULD USE

TMS & Mode Selection

SAP TM — Native mode and carrier optimization engine inside S/4HANA.

Blue Yonder TMS / Oracle OTM — Strong multi-modal rate management and optimization.

Rate Intelligence

Xeneta / Freightos — Ocean and air rate benchmarking platforms.

4 SO WHAT

Choose mode on total landed cost, not on rate. Segment lanes, price the inventory each mode creates, and treat recurring premium freight as a planning failure to fix — not a transport cost to accept.

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